

A Level Computer Science (9618)

Topical Past Paper Worksheet

Topic 8: Databases

Subtopic 8.1: Database Concepts

Concept 8.1.2: Relational Database Terminology

Total Questions: 21

Mark Schemes begin on Page 14

Question 1 | Source: 9618_w25_qp_11 (Q2.d)

- 2 A relational database, WORKEXPERIENCE, stores data about students and the companies where they complete work experience.

Students can complete multiple work experience placements but can only complete one placement at a time.

Students can complete more than one placement at the same company.

Part of the database is shown:

STUDENT(StudentID, FirstName, TelephoneNumber, UniversityName)

PLACEMENT(PlacementID, StudentID, CompanyID, StartDate, EndDate, Complete)

COMPANY(CompanyID, CompanyName, MaxStudentsPerPlacement)

(d) The database tables are repeated here:

STUDENT(StudentID, FirstName, TelephoneNumber, UniversityName)

PLACEMENT(PlacementID, StudentID, CompanyID, StartDate, EndDate, Complete)

COMPANY(CompanyID, CompanyName, MaxStudentsPerPlacement)

Explain what is meant by referential integrity, and how it applies to this database.

.....

.....

.....

.....

.....

..... [3]

Question 2 | Source: 9618_w25_qp_12 (Q4.a)

- 4 A relational database, SHIPPING, stores data about the ships in a company and the containers that are carried on the ships.

The database has the following tables:

CONTAINER(ContainerID, Type, Weight, OwnerName, ShipID)

SHIP(ShipID, Type, Capacity, ShipName)

- (a) Describe the relationship between the two tables. Refer to the primary and foreign keys in your answer.

.....

.....

.....

..... [2]

Question 3 | Source: 9618_s25_qp_11 (Q5.b)

- 5 A company builds and sells furniture to customers. The company stores data about customers, their payment cards and their furniture orders in a database.

The database, FURNITURE, has the following tables:

CUSTOMER(CustomerID, Name, Phone)

CUSTOMER_CARD_DATA(CardID, CustomerID, CardType, CardNumber, EndDate)

ORDER(OrderID, CustomerID, TotalCost, Paid, OrderDate, Complete)

ORDER_ITEM(OrderItemID, OrderID, Type, Height, Width, Depth, Details)

The primary keys are underlined in each table.

The attribute `Complete` in the table `ORDER` stores the Boolean value `TRUE` if the order has been built and `FALSE` if the order has not been built.

- (b) Identify **one** attribute in the table `CUSTOMER_CARD_DATA` that could be a candidate key.

..... [1]

Question 4 | Source: 9618_s25_qp_11 (Q5.c)

- 5 A company builds and sells furniture to customers. The company stores data about customers, their payment cards and their furniture orders in a database.

The database, FURNITURE, has the following tables:

CUSTOMER(CustomerID, Name, Phone)

CUSTOMER_CARD_DATA(CardID, CustomerID, CardType, CardNumber, EndDate)

ORDER(OrderID, CustomerID, TotalCost, Paid, OrderDate, Complete)

ORDER_ITEM(OrderItemID, OrderID, Type, Height, Width, Depth, Details)

The primary keys are underlined in each table.

The attribute `Complete` in the table `ORDER` stores the Boolean value `TRUE` if the order has been built and `FALSE` if the order has not been built.

- (c) Identify **two** tables in the database that contain one or more foreign keys.
Give **one** attribute that is a foreign key in each table.

	Table	Foreign key
1		
2		

[2]

Question 5 | Source: 9618_s25_qp_12 (Q5.b)

- 5 An online game has a database that stores data about users and the characters each user creates in the game. Each user can create multiple characters and purchase multiple items for each character.

The normalised database has the following design:

USER (Username, Password, DateOfBirth)

CHARACTER (CharacterName, CharacterID, Username, Level, Money)

ITEM (ItemName, MinimumLevel, Cost)

CHARACTER_ITEM (CharacterID, ItemName)

- (b) Underline the attribute, or attributes, that form the primary key in each of the tables.

USER (Username, Password, DateOfBirth)

CHARACTER (CharacterName, CharacterID, Username, Level, Money)

ITEM (ItemName, MinimumLevel, Cost)

CHARACTER_ITEM (CharacterID, ItemName)

[2]

Question 6 | Source: 9618_w24_qp_11 (Q2.c)

- 2 A shop repairs electronic devices, for example mobile phones and tablet computers. The shop owner stores the data about the repairs using a file-based approach.

(c) Complete the table by writing a definition for each of the database terms.

Term	Definition
Referential integrity	
Candidate key	
Tuple	

[3]

Question 7 | Source: 9618_w24_qp_12 (Q6.b.i)

- 6 A company uses a relational database to store data about its customers, employees and the individual repair jobs that customers have booked.

(b) The company decides which employees will work on each repair job. An employee can log into the database to access information about their repair jobs.

The database is normalised and includes these tables:

- CUSTOMER stores personal data about each customer
- EMPLOYEE stores personal data about each employee
- LOGIN_DATA stores the username and password for each employee
- JOB stores the data about each repair job
- JOB_EMPLOYEE stores the employees that are working on each repair job.

- [6]

Question 8 | Source: 9618_w24_qp_13 (Q4.a)

- The ice cream is made in batches: a large quantity of one type and flavour of ice cream that is then split into smaller quantities for sale.

Some of the tables in the database are given. The database is not normalised.

CUSTOMER (CustomerID, CompanyName, EmailAddress, TelephoneNumber)

```
SALE(SaleID, BatchID, CustomerID, Quantity, Date)
```

- Foreign key 1

Table name 1

Foreign key 2

Table name 2

Page 6 out of 19

Question 9 | Source: 9618_w24_qp_13 (Q4.d)

4 A company makes ice cream and sells it to shops.

The ice cream is made in batches: a large quantity of one type and flavour of ice cream that is then split into smaller quantities for sale.

The company’s owner has designed a relational database, ICECREAM, to store data about their ice cream and customers.

Some of the tables in the database are given. The database is not normalised.

BATCH(BatchID, Type, Flavour, Size, SellingPrice, EndDate)

CUSTOMER(CustomerID, CompanyName, EmailAddress, TelephoneNumber)

SALE(SaleID, BatchID, CustomerID, Quantity, Date)

(d) Complete the following table by defining each database term.

Database term	Definition
Entity	
Attribute	

[2]

Question 10 | Source: 9618_s24_qp_12 (Q4.a)

4 An assessment board wants to store the marks students achieved in exams in a database named RECORDS.

Part of the database design includes these two tables:

EXAM(ExamID, Subject, Level, TotalMarks)

EXAM_QUESTION(ExamQuestionID, ExamID, QuestionNumber, Question, MaxMark)

(a) Identify the relationship between EXAM and EXAM_QUESTION.

.....
..... [1]

Question 11 | Source: 9618_s24_qp_13 (Q4.a)

- 4 A theatre wants to use a database to store data about the shows that are scheduled, their customers and the seats that the customers have booked.

In the theatre:

- Each show can take place on multiple dates.
- Some dates can have more than one performance.
- There are multiple rows of seats.
- Each seat can be individually booked by its row letter and seat number, for example row E seat 2.

Part of the database design includes these tables:

SHOW(ShowID, Title, Duration)

SEAT(SeatID, RowLetter, SeatNumber)

PERFORMANCE(PerformanceID, ShowID, ShowDate, StartTime)

- (a) Identify the relationship between the tables PERFORMANCE and SHOW.

.....
..... [1]

Question 12 | Source: 9618_w23_qp_12 (Q2.a)

- 2 (a) State what is meant by the following terms in a relational database model.

Entity
.....
.....
Primary key
.....
.....
Referential integrity
.....
..... [3]

Question 13 | Source: 9618_s23_qp_11 (Q2.b.i)

- 2 An organisation uses a database to store data about the types of bird that people have seen.

- (b) The database, `Birds`, stores information about the types of bird and the people who have seen them.

Data about each bird seen is stored with its location and data about the person who saw the bird.

Database `Birds` has the following tables:

`BIRD_TYPE`(`BirdID`, `Name`, `Size`)

`BIRD_SEEN`(`SeenID`, `BirdID`, `Date`, `Location`, `PersonID`)

`PERSON`(`PersonID`, `FirstName`, `LastName`, `EmailAddress`)

- (i) Complete the table by identifying **two** foreign keys and the database table where each is found.

Foreign key	Database table

[2]

Question 14 | Source: 9618_s23_qp_12 (Q2.b)

- 2 A horse riding school uses a database, `Lessons`, to store data about lesson bookings.

This database is created and managed using a Database Management System (DBMS).

- (b) Explain the reasons why referential integrity is important in a database.

.....

.....

.....

.....

.....

..... [3]

Question 15 | Source: 9618_s23_qp_13 (Q4.b)

- 4 A shop rents cars to customers. The shop uses a relational database to store information about the rentals.

(b) Complete the table by writing the missing term or description for each database feature.

Term	Description
.....	An object that data is stored about.
Tuple	
Secondary key	
.....	A field in one table that is linked to a primary key in another table.

[4]

Question 16 | Source: 9618_w22_qp_11 (Q4.c.i)

- 4 A photographer creates a relational database to store data about photographs taken at birthday parties.

The database, `PHOTOGRAPHS`, stores details of the customer, the party, the photographs taken and the cameras used.

The photographer has several cameras that are used for taking the photographs at the parties.

Each camera has a specific lens type (for example, XY32Z) and lighting type (for example, F1672).

Data about each photograph is stored in the database including the party at which it was taken, the time it was taken and the camera used.

The database has these four tables:

```
CUSTOMER(CustomerID, FirstName, LastName, Telephone)
```

```
PARTY(PartyID, CustomerID, PartyDate, StartTime)
```

```
PHOTO_DATA(PhotoID, PartyID, TimeTaken, CameraID)
```

```
CAMERA_DATA(CameraID, LensType, LightingType)
```

(c) The table shows some sample data for the table PHOTO_DATA.

PhotoID	PartyID	TimeTaken	CameraID
ST23-56	BD987	08:34	NIK-02
ST23-57	BD987	08:55	NIK-02
ST23-60	BC08	09:01	CAN-01
ST23-61	BC08	10:23	CAN-12
ST23-62	BC08	10:56	NIK-01

(i) State what is meant by a **tuple**. Give an example of a tuple from PHOTO_DATA.

Tuple

.....

Example

.....

[2]

Question 17 | Source: 9618_w22_qp_12 (Q5.c)

5 A relational database, GARDEN, has the following tables:

OWNER(OwnerID, FirstName, TelephoneNo, TreeID, TreePosition)

TREE(TreeID, ScientificName, MaxHeight, FastGrowing)

(c) State what is meant by a **candidate key** in a relational database.

.....

..... [1]

Question 18 | Source: 9618_w22_qp_13 (Q2.a)

2 The relational database ASTRONOMY is used to store data about telescopes, the companies that own the telescopes and the photographs taken by the telescopes.

The database has these three tables:

COMPANY(TelephoneNumber, CompanyID, CompanyName)

PHOTOGRAPH(PhotoID, TelescopeID, DateTaken, TimeTaken, Elevation)

TELESCOPE(TelescopeID, CompanyID, SerialNumber)

(a) Complete the following table by writing the correct answer for each item.

Item	Answer
a suitable field for the primary key in COMPANY	
a candidate key in TELESCOPE	
the degree of relationship between TELESCOPE and PHOTOGRAPH	

[3]

Question 19 | Source: 9618_w21_qp_11 (Q5.a)

- 5 Javier owns many shops that sell cars. He employs several managers who are each in charge of one or more shops. He uses the relational database CARS to store the data about his business.

Part of the database is shown:

SHOP(ShopID, ManagerID, Address, Town, TelephoneNumber)

MANAGER(ManagerID, FirstName, LastName, DateOfBirth, Wage)

CAR(RegistrationNumber, Make, Model, NumberOfMiles, ShopID)

(a) Tick (✓) **one** box in each row to identify whether each field is a primary key or a foreign key.

Table	Field name	Primary key	Foreign key
MANAGER	ManagerID		
SHOP	ManagerID		
CAR	RegistrationNumber		
CAR	ShopID		

[2]

Question 20 | Source: 9618_s21_qp_11 (Q7.b.i)

- 7 Bobby and Kim are discussing databases.

(b) Bobby has a shop that sells products to customers. His database will store data about his customers, their payment details, orders and the products he sells. Customers will have login details to access their accounts. The database will update customers' payment and login details without keeping any historical records.

(i) Give **one** example of each of the following relationships from Bobby's database.

one-to-one

.....

.....

one-to-many

.....

.....

many-to-many

.....

.....

[3]

Question 21 | Source: 9618_s21_qp_12 (Q1.a)

- 1 Raj owns houses that other people rent from him. He has a database that stores details about the people who rent houses, and the houses they rent. The database, HOUSE_RENTALS, has the following structure:

```
CUSTOMER(CustomerID, FirstName, LastName, DateOfBirth, Email)
HOUSE(HouseID, HouseNumber, Road, Town, Bedrooms, Bathrooms)
RENTAL(RentalID, CustomerID, HouseID, MonthlyCost, DepositPaid)
```

- (a) Give the definition of the following database terms, using an example from the database HOUSE_RENTALS for each definition.

Term	Definition and example
Field	
Entity	
Foreign key	

[6]

MARK SCHEMES

Mark Scheme for Question 1 | Source: 9618_w25_ms_11 (Q2.d)

2(d)	1 mark per bullet point, max 3 marks Generic mark points (max 2) • Referential integrity ensures that related data is consistent • Referential integrity ensures that every foreign key has a corresponding primary key • Referential integrity provides for cascading update / delete • Referential integrity ensures that if a primary key is deleted or modified all linked records in foreign table will be deleted or modified // Referential integrity stops "orphaned records"—records in a table that point to an entry in another table that no longer exists Specific mark points (max 2) for good examples e.g. • CompanyID is a foreign key in PLACEMENT table and is dependent on the primary key CompanyID in COMPANY table • If a record is deleted from the STUDENT table, then all records with the same StudentID will be deleted from the PLACEMENT table • If a CompanyID is modified in the company table, all records with that CompanyID in the placement table will also be modified	3
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Mark Scheme for Question 2 | Source: 9618_w25_ms_12 (Q4.a)

4(a)	1 mark per bullet point, max 2 marks • The relationship between SHIP and CONTAINER is one-to-many (1:M) • The primary key <u>ShipID</u> in the SHIP table is linked to the foreign key ShipID in the CONTAINER table	2
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Mark Scheme for Question 3 | Source: 9618_s25_ms_11 (Q5.b)

5(b)	1 mark for: CardNumber	1
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Mark Scheme for Question 4 | Source: 9618_s25_ms_11 (Q5.c)

5(c)

1 mark per row to max 2

2

Table	Foreign key
ORDER_ITEM	OrderID
ORDER	CustomerID
CUSTOMER_CARD_DATA	CustomerID

Mark Scheme for Question 5 | Source: 9618_s25_ms_12 (Q5.b)

5(b)	1 mark for the three single Primary Keys 1 mark for the composite key USER(<u>Username</u> , Password, DateOfBirth) CHARACTER(CharacterName, <u>CharacterID</u> , Username, Level, Money)	2
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Mark Scheme for Question 6 | Source: 9618_w24_ms_11 (Q2.c)

2(c)	1 mark for each correct definition	3								
<table><tr><th>Database Term</th><th>Definition</th></tr><tr><td>Referential Integrity</td><td>All duplicate entries of data between tables are consistent // all foreign keys are matched to an appropriate primary key</td></tr><tr><td>Candidate Key</td><td>A field that could be a primary key but is not // an attribute or smallest set of attributes in a table where no tuple has the same value</td></tr><tr><td>Tuple</td><td>A row / record in a table // one instance of an entity in a table</td></tr></table>			Database Term	Definition	Referential Integrity	All duplicate entries of data between tables are consistent // all foreign keys are matched to an appropriate primary key	Candidate Key	A field that could be a primary key but is not // an attribute or smallest set of attributes in a table where no tuple has the same value	Tuple	A row / record in a table // one instance of an entity in a table
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Candidate Key	A field that could be a primary key but is not // an attribute or smallest set of attributes in a table where no tuple has the same value									
Tuple	A row / record in a table // one instance of an entity in a table									

Mark Scheme for Question 7 | Source: 9618_w24_ms_12 (Q6.b.i)

6(b)(i)	1 mark each to max 6 - CUSTOMER to JOB is 1 to many ... implemented by Primary Key in CUSTOMER is Foreign Key in JOB - EMPLOYEE to LOGIN_DATA is 1 to 1 ... implemented by Primary Key in EMPLOYEE is Foreign Key in LOGIN_DATA - JOB to JOB_EMPLOYEE is 1 to many ... implemented by Primary Key in JOB is Foreign Key in JOB_EMPLOYEE - EMPLOYEE to JOB_EMPLOYEE is 1 to many ... implemented by Primary Key in EMPLOYEE is Foreign Key in JOB_EMPLOYEE	6
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Mark Scheme for Question 8 | Source: 9618_w24_ms_13 (Q4.a)

4(a)	1 mark each: - Foreign key: BatchID, table BATCH - Foreign key: CustomerID, table CUSTOMER	2
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Mark Scheme for Question 9 | Source: 9618_w24_ms_13 (Q4.d)

4(d)	1 mark for each definition Entity: - A real-life object that is represented as a table Attribute: - An item of data about an entity	2
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Mark Scheme for Question 10 | Source: 9618_s24_ms_12 (Q4.a)

4(a)	1 mark for: 1-to-many	1
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Mark Scheme for Question 11 | Source: 9618_s24_ms_13 (Q4.a)

4(a)	1 mark for: many to 1 // there are many performances of each show	1
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Mark Scheme for Question 12 | Source: 9618_w23_ms_12 (Q2.a)

2(a)	1 mark for each term (max 3) Entity: - An object about which data can be stored Primary key: - The unique attribute / combination of attributes used to identify the record / tuple Referential integrity: - Makes sure that if data is changed in one place the change is reflected in all related records - cascading update/delete - Makes sure that data that does not exist cannot be referenced - Ensures that every foreign key has a corresponding primary key // A logical dependency of a foreign key on a primary key - Ensures that the data in the database is consistent / up to date - Prevents records from being added/deleted/modified incorrectly - Makes sure any queries return accurate and complete results	3
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Mark Scheme for Question 13 | Source: 9618_s23_ms_11 (Q2.b.i)

2(b)(i)	1 mark for each field name and table	2						
	<table><tr><th>Foreign key</th><th>Database table</th></tr><tr><td>BirdID</td><td>BIRD_SEEN</td></tr><tr><td>PersonID</td><td>BIRD_SEEN</td></tr></table>	Foreign key	Database table	BirdID	BIRD_SEEN	PersonID	BIRD_SEEN	
Foreign key	Database table							
BirdID	BIRD_SEEN							
PersonID	BIRD_SEEN							

Mark Scheme for Question 14 | Source: 9618_s23_ms_12 (Q2.b)

2(b)	1 mark each to max 3 - Referential Integrity makes sure data is consistent - Referential Integrity makes sure all data is up-to-date - Referential integrity ensures that every foreign key has a corresponding primary key - Referential Integrity prevents records from being added / deleted / modified incorrectly - Referential Integrity makes sure that if data is changed in one place the change is reflected in all related records - Referential Integrity makes sure any queries return accurate and complete results	3
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Mark Scheme for Question 15 | Source: 9618_s23_ms_13 (Q4.b)

4(b)

1 mark each

4

Term	Description
Entity	An object that data is stored about.
Tuple	A row of data in a table about one instance of an object.
Secondary key	An additional/alternative key used as well as the primary key to locate specific data // a candidate key that has not been chosen as a primary key.
Foreign key	A field in one table that is linked to a primary key in another table.

Mark Scheme for Question 16 | Source: 9618_w22_ms_11 (Q4.c.i)

4(c)(i)	1 mark for the definition, 1 mark for the example: - definition: a single row in a table - example: from the PHOTO_DATA table	2
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Mark Scheme for Question 17 | Source: 9618_w22_ms_12 (Q5.c)

5(c)	1 mark for: An attribute / field (or set of attributes / fields) that could be a primary key	1
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Mark Scheme for Question 18 | Source: 9618_w22_ms_13 (Q2.a)

2(a)	1 mark for each correct answer.	3								
<table><tr><th>Item</th><th>Answer</th></tr><tr><td>a suitable field for the primary key in COMPANY</td><td>CompanyID</td></tr><tr><td>a candidate key in TELESCOPE</td><td>SerialNumber // TelescopeID</td></tr><tr><td>the degree of relationship between TELESCOPE and PHOTOGRAPH</td><td>1:M / 1 to many</td></tr></table>			Item	Answer	a suitable field for the primary key in COMPANY	CompanyID	a candidate key in TELESCOPE	SerialNumber // TelescopeID	the degree of relationship between TELESCOPE and PHOTOGRAPH	1:M / 1 to many
Item	Answer									
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a candidate key in TELESCOPE	SerialNumber // TelescopeID									
the degree of relationship between TELESCOPE and PHOTOGRAPH	1:M / 1 to many									

Mark Scheme for Question 19 | Source: 9618_w21_ms_11 (Q5.a)

5(a)	1 mark for 2 or 3 correct ticks, 2 marks for 4 correct ticks	2
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Mark Scheme for Question 20 | Source: 9618_s21_ms_11 (Q7.b.i)

7(b)(i)	1 mark for each correct example one-to-one - e.g. customer to payment details // customer to login details one-to-many - e.g. customer to order many-to-many - e.g. order to product // customer to product	3
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Mark Scheme for Question 21 | Source: 9618_s21_ms_12 (Q1.a)

1(a)	1 mark for definition, 1 mark for appropriate example in each	6								
	<table><tr><th>Term</th><th>Definition and example</th></tr><tr><td>Field</td><td>A column/attribute in a table e.g. CustomerID in the table CUSTOMER</td></tr><tr><td>Entity</td><td>Anything that data can be stored about e.g. A customer or a house</td></tr><tr><td>Foreign Key</td><td>A field in one table that is linked to a Primary Key in another table e.g. CustomerID / HouseID <u>in table RENTAL</u></td></tr></table>	Term	Definition and example	Field	A column/attribute in a table e.g. CustomerID in the table CUSTOMER	Entity	Anything that data can be stored about e.g. A customer or a house	Foreign Key	A field in one table that is linked to a Primary Key in another table e.g. CustomerID / HouseID <u>in table RENTAL</u>	
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